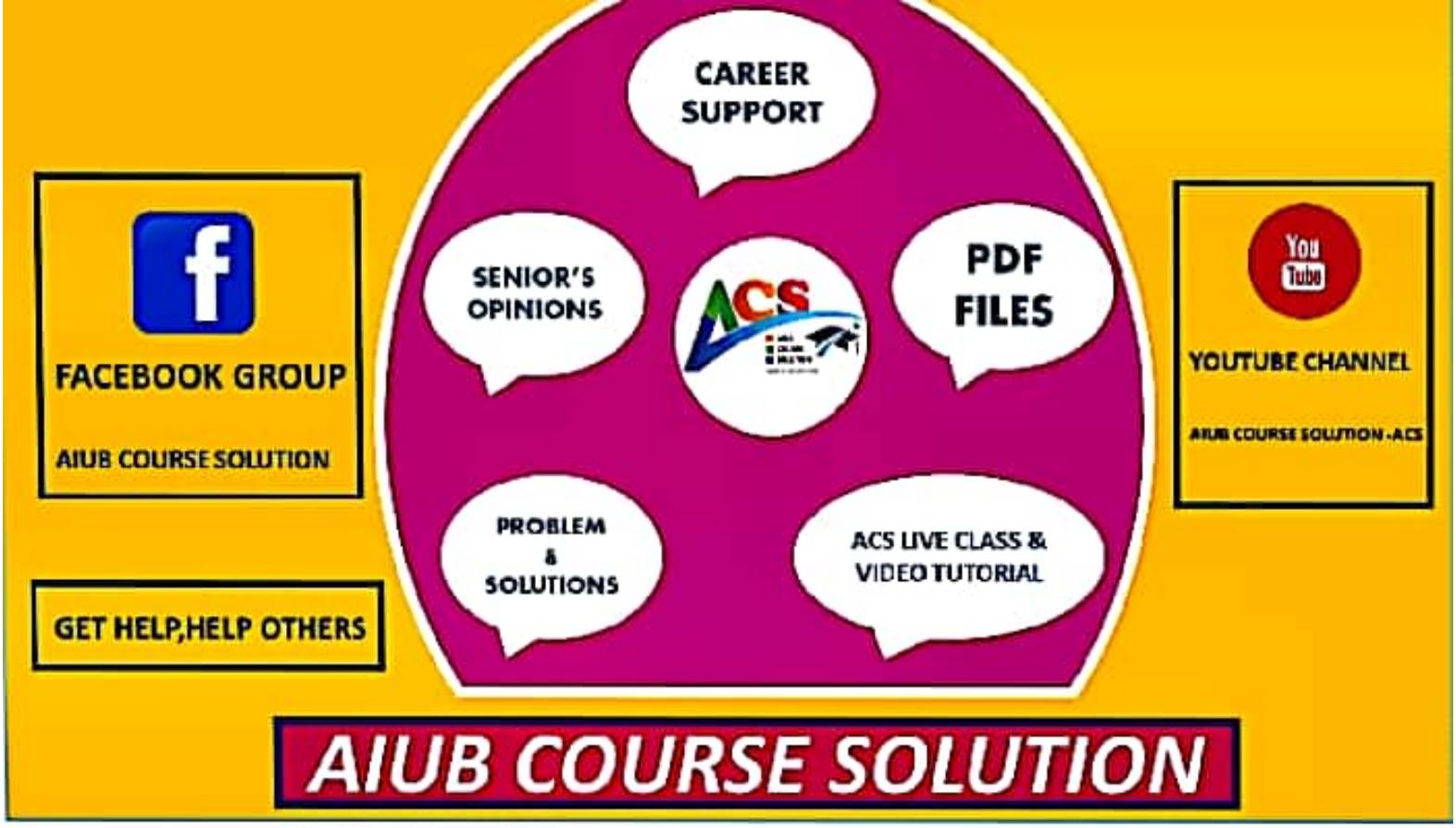


TOGETHER WE CAN ACHIEVE MORE MOREMORE

COURSE NAME: **OOAD**

CHAPTER:

SOLVED BY

NAME: **Ismath Jahan**

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Model

⊛ modeling:-

- Modeling is a proven and well accepted engineering technique.
- We do model so that we can better understand the system we are developing.
- A model is a simplification of reality.
- A model provides the blueprints of a system.
- We achieve four aims by using modelling—
  1. Visualize a system.
  2. Specify the structure.
  3. template that constructing a system.
  4. model document the decissions we have made.

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* Principal of modeling –

- Choose models intelligently.
- Every model may be expressed at different levels of precision.
- The best models are connected to reality.
- No single model is sufficient.

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Uml

* UML → Unified modeling language
  ↓
  Omg standard
  (Object management group)

- UML is a modeling language to express and design documents, software.
- Not a process.
- Independent of implementation language.

* Graphical notation for —
  - Specifying.
  - Visualizing.
  - Constructing.
  - Documenting.

- Increase communication of product to customers and developers.

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• Building blocks of UML –

– 3 kinds of blocks.

① Things ② Relationships. ③ Diagrams.

• Things in the UML

– Structural things → static parts of a model.
– Behavioral things → dynamic " " " " .
– Grouping things → Organizational " " " .
– Annotation things → Explanatory " " " " .

⊛ Relationship in the UML:

– 4 kinds of Relationship.

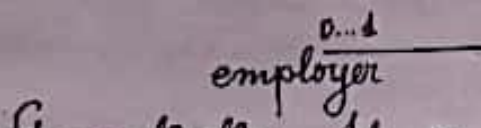
① Dependency: It is a semantic relationship between two things are dependent with one another.

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② Association : It is a structural relationship that describes a set of links.



③ Generalization : It is a specialization relation -ship. An actor can inherit the role of other actor. So that the child shares the behaviour and behavior of the parent.



④ Realization : It is a semantic relationship between classifiers.



⑤ Diagram in the UML :

— A diagram is a graphical presentation of a set of elements.

— There are 9 types of diagrams in UML.

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1. Class diagram: It shows a set of classes, interfaces & collaborations and their relationships.

2. Use case diagram: It shows a set of use cases & actors and their relationships. It addresses static use case view of a system.

* There are three types of modeling in UML.

(1) Structural modeling — It shows the static parts of a system. It consists diagram.

Example — class diagram, object diagram. This model represents the framework for the system. It never describes the dynamic behaviour of the system.

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② Behavioral modeling:

It shows the dynamic parts of a system.
It consists of the following diagrams:

- Activity diagrams.
- Use case diagrams.
- Interaction diagrams.

- This diagrams show the dynamic sequence of flow in the system.

③ Architectural modeling:

It represents the overall framework of the system. It is a combination of structural & behavioral elements of the system. It is the blueprint for the entire system. package diagram is used.

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